

Mohd Farhan

9354378146 | mohdfarhan4002@gmail.com | LinkedIn | Portfolio | Github — Delhi, India

SUMMARY

Results-driven Computer Science Engineering student with hands-on experience in AI-driven solutions. Proficient in Python and Java with a strong foundation in Data Structures and Algorithms. Demonstrated ability to design, develop, and deploy scalable applications in fast-paced hackathon environments. Adept at problem-solving, research, and cross-functional collaboration, and eager to contribute technical expertise to build impactful AI-powered systems.

TECHNICAL SKILLS

Programming Languages: Python, Javascript, Java, React, C, SQL, Dart

ML / AI-: TensorFlow, PyTorch, SHAP, MLflow, Streamlit

Libraries & Tools: NumPy, Pandas, Scikit-learn, OpenCV, NLTK, Git, Github, Docker

PROJECTS

Movie Maverick — AI Movie Recommendation Platform [GitHub](#) *Python • Flask • PostgreSQL • Redis • Docker • Scikit-learn • Gemini AI*

- Engineered a hybrid recommendation engine combining user-based collaborative filtering (k-NN), content-based filtering, and LLM-powered suggestions via the Google Gemini API to improve recommendation quality.
- Built a full-stack production application with authentication, social features (follow system, reviews, activity feed), containerised deployment via Docker Compose, Nginx reverse proxy, and a Redis-based caching layer.
- Implemented robust security controls including rate limiting, CSRF protection, XSS mitigation (Bleach), input validation with Marshmallow schemas, and secure password storage using bcrypt hashing.

Advanced Chess Platform [GitHub](#) *TypeScript • React • Node.js • WebSockets • PostgreSQL • Redis • Stockfish*

- Developed a real-time multiplayer chess platform with WebSocket-based matchmaking, Stockfish-powered AI opponent, tournament formats (Swiss, Round-Robin, Arena), and a puzzle trainer with Elo-based ratings.
- Architected as a Turborepo monorepo with separate Express REST API, WebSocket server, and React frontend; used Prisma ORM with PostgreSQL for persistence and Redis for move queuing and caching.
- Delivered social features (friends, messaging, clubs), advanced game analytics (opening performance, rating progression, accuracy scores), an achievement system, and 50+ documented REST API endpoints.

Customer Churn Intelligence System [GitHub](#) *Python • Scikit-learn • SHAP • Pandas • NumPy • Random Forest*

- Built an end-to-end ML pipeline for telecom customer churn prediction, including EDA, feature engineering, Random Forest modelling, hyperparameter tuning, and model evaluation.
- Applied Explainable AI techniques with SHAP to surface key churn drivers (e.g., contract type, tenure) and translated findings into concrete customer retention strategy recommendations.

Emotion Recognition System [GitHub](#) *Python • TensorFlow • Keras • OpenCV • CNN • Deep Learning*

- Designed and trained a Convolutional Neural Network (CNN) to classify seven facial emotions (happy, sad, angry, fear, surprise, disgust, neutral) from real-time webcam input using OpenCV.
- Improved validation performance through data augmentation, dropout regularisation, and transfer learning; deployed the model as an interactive Streamlit web application for real-time inference.

EDUCATION

IILM University
B.Tech in Computer Science, with Data Science
Good Samaritan School
12th Class

Greater Noida, U.P
Aug 2023 – May 2027
New Delhi
2021-22